Second Order ODEs

The **general form** of a second order ODE is

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = f(t, y, \frac{\mathrm{d}y}{\mathrm{d}t})$$

for some given function f.

1 Linear Homogeneous Equations

Definition 1.1 (Linear Equation).
A second order ODE is called **linear** if it can be written in the form

$$a(t) \cdot y'' + b(t) \cdot y' + c(t) \cdot y = g(t)$$

If $a(t) \neq 0$, we can divide both sides by $a(t)$ to get $y'' + p(t) \cdot y' + q(t) \cdot y = r(t)$.

Definition 1.2 (Homogeneous Equation).
A linear equation is called **homogeneous** if it can be written in the form

$$a(t) \cdot y'' + b(t) \cdot y' + c(t) \cdot y = 0$$

Otherwise, it is called **non-homogeneous**.

Theorem 1.1 (Existence and Uniqueness Theorem).
Consider the IVP:

$$y'' + p(t) \cdot y' + q(t) \cdot y = r(t), \qquad y(t_0) = y_0, \quad y'(t_0) = y_1$$

if there exists an open interval $I \subseteq \mathbb{R}$ such that $t_0 \in I$ and p, q and r are **continuous** on I, then there is a **exact one** solution $y = \phi(t)$, and it exists **throughout** I.

Theorem 1.2 (Principle of Superposition).
If y_1 and y_2 are two solutions of the **homogeneous equation**

$$a(t) \cdot y'' + b(t) \cdot y' + c(t) \cdot y = 0$$

then any linear combination $c_1 y_1(t) + c_2 y_2(t)$ is also a solution of the ODE.

Definition 1.3 (Wronskian).
The **Wronskian**, or **Wronskian determinant**, of two functions y_1 and y_2 is defined as

$$W(y_1, y_2)(t) := \begin{vmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{vmatrix} = y_1(t) \cdot y_2'(t) - y_2(t) \cdot y_1'(t)$$

Theorem 1.3 (Abel's Theorem).
Suppose $p(t)$ and $q(t)$ are continuous on an open interval I, and y_1 and y_2 are two non-zero solutions of the ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \tag{1}$$

Then, the Wronskian of y_1 and y_2 is given by

$$W(y_1, y_2)(t) = c \cdot \exp\left(-\int p(t)\,\mathrm{d}t\right) \qquad \text{where } c \in \mathbb{R} \text{ is a constant depending on } y_1, y_2$$

Remark. If $y(t), p(t), q(t) \in \mathbb{C}$, then $c \in \mathbb{C}$.

Corollary $\quad \exists t_0 \in I \text{ s.t. } W(y_1, y_2)(t_0) \neq 0 \implies \forall t \in I,\ W(y_1, y_2)(t) \neq 0$

Proof (Abel's Theorem).

$$\begin{cases} y_2 y_1'' + y_2 p y_1' + y_2 q y_1 = 0 \\ y_1 y_2'' + y_1 p y_2' + y_1 q y_2 = 0 \end{cases} \implies (y_2 y_1'' - y_1 y_2'') + p \cdot (y_2 y_1' - y_1 y_2') = 0$$
$$\implies W' + p \cdot W = 0$$
$$\implies |W(t)| = \exp\left(-\int p(t)\,\mathrm{d}t\right) \quad \text{or} \quad W(t) = 0$$

□

Theorem 1.4 (Linear Dependence and Wronskian).
Suppose y_1 and y_2 are two solutions of an ODE in the form of eq. (1). Then,

$$y_1(t) \quad \text{and} \quad y_2(t) \quad \text{are } \textbf{linearly independent} \text{ on} \quad t \in I \qquad \iff \qquad \forall t \in I,\ W(y_1, y_2)(t) \neq 0$$

Proof.

$$\begin{aligned}
\text{linear independence} \quad &\iff \quad \forall c_1, c_2 \in \mathbb{C},\ \Big(c_1 y_1(t) + c_2 y_2(t) = 0\Big) \to \Big(c_1 = c_2 = 0\Big) && \text{[Proposition 1]} \\
&\iff \quad \forall c_1, c_2 \in \mathbb{C},\ \begin{bmatrix} y_1(t) & y_2(t) \\ y_1'(t) & y_2'(t) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \to \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} && \text{[Proposition 2]} \\
&\iff \quad W(y_1, y_2)(t) \not\equiv 0 \\
&\iff \quad \forall t \in I,\ W(y_1, y_2)(t) \neq 0 \quad \text{by the corollary of Abel's Theorem}
\end{aligned}$$

([Proposition 2] $\implies$ [Proposition 1]) is obtained by combining [Proposition 2] with the following fact:

$$\forall c_1, c_2 \in \mathbb{C},\ \Big(c_1 y_1(t) + c_2 y_2(t) = 0\Big) \to \Big(c_1 y_1'(t) + c_2 y_2'(t) = 0\Big)$$

□

Theorem 1.5.
Suppose $p(t)$ and $q(t)$ are continuous on an open interval I, and y_1 and y_2 are two solutions of the ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \quad \text{for} \quad t \in I \tag{2}$$

Let $\phi(t)$ be the **unique** solution (by Theorem 1.1) of the IVP:

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \qquad y(t_0) = m, \quad y'(t_0) = n \quad \text{for} \quad t \in I \quad \text{and} \quad t_0 \in I \tag{3}$$

Then, for $t \in I$,

$$\exists!\, c_1, c_2 \in \mathbb{R} \text{ s.t. } \phi(t) = c_1 y_1(t) + c_2 y_2(t) \qquad \iff \qquad W(y_1, y_2)(t_0) \neq 0 \tag{4}$$

Remark. In the assumption, y_1 and y_2 are not required to be linearly independent. Thus c_1, c_2 are required to be unique in the left hand side of eq. (4).

Proof.

Sufficiency. Since $\phi(t_0) = m$ and $\phi'(t_0) = n$, we have

$$\begin{aligned}
\exists!\, c_1, c_2 \in \mathbb{R} \text{ s.t. } \phi(t) = c_1 y_1(t) + c_2 y_2(t) \quad &\implies \quad \exists!\, c_1, c_2 \in \mathbb{R} \ \text{ s.t.} \begin{bmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} m \\ n \end{bmatrix} \\
&\implies \quad \exists b_1, b_2 \in \mathbb{R},\ \exists!\, c_1, c_2 \in \mathbb{R} \ \text{ s.t.} \begin{bmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} \\
&\iff \quad W(y_1, y_2)(t_0) \neq 0
\end{aligned}$$

Necessity.

$$W(y_1, y_2)(t_0) \neq 0 \quad \implies \quad \exists!\, c_1, c_2 \in \mathbb{R} \ \text{ s.t.} \begin{bmatrix} y_1(t_0) & y_2(t_0) \\ y_1'(t_0) & y_2'(t_0) \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} m \\ n \end{bmatrix}$$

Let $\varphi(t) := c_1 y_1(t) + c_2 y_2(t), t \in I$. Then we have $\varphi(t_0) = m$ and $\varphi'(t_0) = n$.

$$\begin{aligned}
\text{Theorem 1.2 (Principle of Superposition)} \quad &\implies \quad \varphi(t) \text{ is a solution of the ODE eq. (2).} \\
\text{Theorem 1.1 (Existence and Uniqueness Theorem)} \quad &\implies \quad \varphi(t) \text{ is the } \textbf{only} \text{ solution of the IVP in eq. (3).}
\end{aligned}$$

Thus, $\varphi(t) = \phi(t)$ and we have

$$W(y_1, y_2)(t_0) \neq 0 \quad \Longrightarrow \quad \exists!\, c_1, c_2 \in \mathbb{R} \text{ s.t. } \phi(t) = \varphi(t) = c_1 y_1(t) + c_2 y_2(t)$$

□

Theorem 1.6.
Suppose $p(t)$ and $q(t)$ are continuous on an open interval I, and y_1 and y_2 are two solutions of ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0$$

for $t \in I$. Then,

The **general solution** is given by $\phi(t) = c_1 y_1(t) + c_2 y_2(t)$. $\iff$ $\exists t_0 \in I$ s.t. $W(y_1, y_2)(t_0) \neq 0$.

y_1, y_2 are said to form a **fundamental set of solutions (FSS)** of the ODE, if their Wronskian is non-zero.

Remark. Theorem 1.6 is used to determine whether a given pair of solutions y_1, y_2 form a FSS, rather than verifying the existence and form of solutions. An ODE satisfying the condition (continuous coefficients) always has solutions, and the solutions form a 2D vector space. By Theorem 1.4 (Linear Dependence and Wronskian), the linear independence of y_1, y_2 also indicates that they form a FSS.

Proof.
Let S denote the set of all solutions of the ODE. We aim to prove that

$$\forall y \in S, \exists! c_1, c_2 \in \mathbb{R} \text{ s.t. } y(t) = c_1 y_1(t) + c_2 y_2(t) \quad \leftrightarrow \quad \exists t_0 \in I \text{ s.t. } W(y_1, y_2)(t_0) \neq 0$$

Let $\mathcal{S}_{\texttt{IVP}}[t^*, m, n]$ denote the solution of the following IVP (by Theorem 1.1, there is exactly one solution):

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \qquad y(t^*) = m, \quad y'(t^*) = n \quad \text{for} \quad t \in I \quad \text{and} \quad t^* \in I$$

By Theorem 1.5, we have

$$\forall t_0 \in I, \forall m, n \in \mathbb{R}\Big[\Big(W(y_1, y_2)(t_0) \neq 0\Big) \leftrightarrow \Big(\exists! c_1, c_2 \in \mathbb{R} \text{ s.t. } \mathcal{S}_{\texttt{IVP}}[t_0, m, n] = c_1 y_1 + c_2 y_2\Big)\Big]$$

Instantiate m, n with $y(t_0), y'(t_0)$ for all $y \in S$:

$$\forall t_0 \in I, \forall y \in S\Big[\Big(W(y_1, y_2)(t_0) \neq 0\Big) \leftrightarrow \Big(\exists! c_1, c_2 \in \mathbb{R} \text{ s.t. } \mathcal{S}_{\texttt{IVP}}[t_0, y(t_0), y'(t_0)] = c_1 y_1 + c_2 y_2\Big)\Big]$$

Since y satisfies the ODE and initial conditions itself, $\mathcal{S}_{\texttt{IVP}}[t_0, y(t_0), y'(t_0)]$ is identical to y. Thus we obtain

$$\forall t_0 \in I, \forall y \in S\Big[\Big(W(y_1, y_2)(t_0) \neq 0\Big) \leftrightarrow \Big(\exists! c_1, c_2 \in \mathbb{R} \text{ s.t. } y = c_1 y_1 + c_2 y_2\Big)\Big] \tag{5}$$

Now apply the following two corollaries from first-order formal logic:

$$\Big(\exists x. R(x)\Big) \wedge \forall x. \Big(R(x) \to (P(x) \leftrightarrow Q)\Big) \quad \Longrightarrow \quad \Big(\exists x. (R(x) \wedge P(x))\Big) \leftrightarrow Q$$
$$\Big(\exists x. R(x)\Big) \wedge \forall x. \Big(R(x) \to (P(x) \leftrightarrow Q)\Big) \quad \Longrightarrow \quad \Big(\forall x. (R(x) \to P(x))\Big) \leftrightarrow Q$$

Since both I and S are non-empty, we can transform eq. (5) into

$$\Big(\exists t_0 \in I,\ W(y_1, y_2)(t_0) \neq 0\Big) \leftrightarrow \forall y \in S\Big(\exists! c_1, c_2 \in \mathbb{R} \text{ s.t. } y = c_1 y_1 + c_2 y_2\Big)$$

This completes the proof. □

Theorem 1.7 (Existence of Fundamental Set of Solutions)**.**
If $p(t)$ and $q(t)$ are continuous on an open interval I, then there must exist fundamental sets of solutions for the following ODE:

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \quad \text{for} \quad t \in I \tag{6}$$

A FSS can be constructed as follows: Pick any $t_0 \in I$, and let y_1 be the solution of eq. (6) with initial conditions

$$y_1(t_0) = 1, \quad y_1'(t_0) = 0$$

Let y_2 be the solution of eq. (6) with initial conditions

$$y_2(t_0) = 0, \quad y_2'(t_0) = 1$$

Then, y_1 and y_2 form a FSS of eq. (6).

Proof.
Theorem 1.1 (Existence and Uniqueness Theorem) guarantees the existence of y_1 and y_2. By the definition of Wronskian, we have

$$W(y_1, y_2)(t_0) = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \neq 0$$

By the corollary of Abel's Theorem, $W(y_1, y_2)(t) \neq 0$ for all $t \in I$. y_1 and y_2 form a FSS of eq. (6). ☐

2 Homogeneous Equations with Constant Coefficients

2.1

Theorem 2.1.
Given an ODE with **real-valued** coefficients

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0$$

if y is a **complex-valued** solution of the ODE, then $\mathrm{Re}(y), \mathrm{Im}(y)$ are also solutions of the ODE.

Proof.
Substitute $y(t) = u(t) + \mathrm{i}\, v(t)$ into the ODE, and it can be verified that both $u(t)$ and $v(t)$ satisfy the ODE. ☐

Theorem 2.2.
For the linear ODE with constant coefficients

$$ay'' + by' + cy = 0 \tag{7}$$

the **characteristic equation** is given by

$$ar^2 + br + c = 0$$

Let r_1, r_2 be the two roots and $\Delta := b^2 - 4ac$ be the discriminant. Then,

- If $\Delta > 0$, then $r_1 \neq r_2 \in \mathbb{R}$, and the general solution is given by

$$y(t) = c_1 e^{r_1 t} + c_2 e^{r_2 t}$$

- If $\Delta = 0$, then $r_1 = r_2 \in \mathbb{R}$, and the general solution is given by

$$y(t) = (c_1 + c_2 t) e^{-\frac{b}{2a} t}$$

- If $\Delta < 0$, then $r_1 = \overline{r_2} \in \mathbb{C}$. Let $\lambda := \mathrm{Re}(r_1)$ and $\mu := |\mathrm{Im}(r_1)|$. The general solution is given by

$$y(t) = e^{\lambda t}\Big(c_1 \cos(\mu t) + c_2 \sin(\mu t)\Big) \qquad \text{(real-valued when } a, b, c, c_1, c_2 \in \mathbb{R})$$
$$\text{or} \quad y(t) = e^{\lambda t}\Big(c_1 \cos(\mu t) + \mathrm{i}\, c_2 \sin(\mu t)\Big) \qquad \text{(complex-valued)}$$

Proof.
The general solutions are derived by substituting the ansatz $y(t) = e^{rt}$ into the differential equation and applying Theorem 1.6.

When $\Delta < 0$, the characteristic equation yields complex conjugate roots, leading to complex-valued solutions. However, if the coefficients $a, b, c \in \mathbb{R}$, we may apply Theorem 2.1 to construct an equivalent real-valued general solution.

When $\Delta = 0$, he characteristic equation has a repeated real root r, yielding only one solution $y_1(t) = e^{rt}$. To form a fundamental set of solutions, we require a second linearly independent solution. Using Theorem 1.3 (Abel's Theorem), we can obtain $y_2(t) = te^{rt}$. *(Note: Any solution linearly independent of $y_1(t)$ is acceptable, though te^{rt} is the standard choice.)* ☐

Theorem 2.3 (Reduction of Order)**.**
If y_1 is a non-zero solution of the ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0 \quad \text{for} \quad t \in I$$

then a second linearly independent solution y_2 can be found by

$$y_2(t) = y_1(t) \int \frac{\exp\left(-\int p(t)\,\mathrm{d}t\right)}{\left[y_1(t)\right]^2}\,\mathrm{d}t$$

Proof.
We suppose $y_2(t) = v(t)y_1(t)$ for some function v. Substituting y_2 into the ODE, we have

$$y_1 v'' + (2y_1' + py_1)v' = 0$$

Thus v can be solved.

$$v'(t) = \frac{1}{y_1^2(t)} \exp\left(-\int p(t)\,\mathrm{d}t\right)$$

□

2.2 Euler Equations

Definition 2.1 (Euler Equation)**.**
Euler equations are the differential equations of the form

$$x^2 \frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + Ax\frac{\mathrm{d}y}{\mathrm{d}x} + By = 0 \quad \text{for} \quad x > 0 \tag{8}$$

where A, B are constants.

Let $x = e^t$, then we have

$$\begin{aligned}\frac{\mathrm{d}y}{\mathrm{d}x} &= \frac{\mathrm{d}y}{\mathrm{d}(e^t)} \\ &= \frac{1}{e^t} \cdot \frac{\mathrm{d}y}{\mathrm{d}t}\end{aligned}$$

$$\begin{aligned}\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} &= \frac{\mathrm{d}}{\mathrm{d}(e^t)}\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right) \\ &= \frac{1}{e^t\,\mathrm{d}t} \cdot \mathrm{d}\left(\frac{1}{e^t}\frac{\mathrm{d}y}{\mathrm{d}t}\right) \\ &= \frac{1}{e^t\,\mathrm{d}t}\left[-\frac{\mathrm{d}t}{e^t}\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{1}{e^t}\,\mathrm{d}\left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)\right] \\ &= \frac{1}{e^{2t}}\left[-\frac{\mathrm{d}y}{\mathrm{d}t} + \frac{\mathrm{d}^2 y}{\mathrm{d}t^2}\right]\end{aligned}$$

Thus,

$$x\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y}{\mathrm{d}t} \qquad\qquad x^2\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} = \frac{\mathrm{d}^2 y}{\mathrm{d}t^2} - \frac{\mathrm{d}y}{\mathrm{d}t}$$

Substituting the above into eq. (8), we have

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} + (A-1)\frac{\mathrm{d}y}{\mathrm{d}t} + By = 0$$

which is a linear homogeneous ODE with constant coefficients.

3 Non-Homogeneous Equations

Theorem 3.1.
Suppose $p(t), q(t), r(t)$ are continuous on an interval I. Then the **general solution** of the ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = r(t)$$

is given by

$$y(t) = y_h(t) + y_p(t)$$

where y_h, or denoted as y_c and called the **complementary solution**, is the general solution of the corresponding homogeneous ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = 0$$

and y_p is a **particular solution** of the non-homogeneous ODE.

3.1 Method of Undetermined Coefficients

Consider the non-homogeneous ODE with constant coefficients

$$ay'' + by' + cy = r(t)$$

When the non-homogeneous term $r(t)$ is of a specific form, we can apply the method of undetermined coefficients to find a particular solution y_p. And since $ay'' + by' + cy = 0$ can be easily solved, we can obtain the general solution of the non-homogeneous ODE.

$r(t)$	$y_p(t)$
$a_n t^n + \cdots + a_1 t + a_0$	$t^s(A_n t^n + \cdots + A_1 t + A_0)$
$(a_n t^n + \cdots + a_1 t + a_0)e^{\alpha t}$	$t^s(A_n t^n + \cdots + A_1 t + A_0)e^{\alpha t}$
$(a_n t^n + \cdots + a_1 t + a_0)e^{\alpha t}\sin \beta t$ or $(a_n t^n + \cdots + a_1 t + a_0)e^{\alpha t}\cos \beta t$	$t^s\Big[(A_n t^n + \cdots + A_1 t + A_0)\sin \beta t + (B_n t^n + \cdots + B_1 t + B_0)\cos \beta t\Big]e^{\alpha t}$

Notes. Here, s is the smallest non-negative integer ($s \in \{0, 1, 2\}$) ensuring that $y_p(t)$ and $y_h(t)$ possess no common terms. Equivalently, s represents the multiplicity of the root 0, α, or $\alpha + \mathrm{i}\beta$ in the characteristic equation, corresponding to the respective forms of the non-homogeneous term $r(t)$.

Theorem 3.2.
If y_1 is a solution to $ay'' + by' + cy = r_1(t)$ and y_2 is a solution to $ay'' + by' + cy = r_2(t)$, then $y_1 + y_2$ is a solution to the equation $ay'' + by' + cy = r_1(t) + r_2(t)$.

3.2 Variation of Parameters

Theorem 3.3 (Variation of Parameters)**.**
Suppose $p(t), q(t), r(t)$ are continuous on an open interval I. Then a particular solution of the ODE

$$y'' + p(t) \cdot y' + q(t) \cdot y = r(t) \quad \text{for} \quad t \in I \tag{9}$$

can be found by

$$y_p(t) = u_1(t)y_1(t) + u_2(t)y_2(t) \tag{10}$$

where (y_1, y_2) is a fundamental set of solutions of the corresponding homogeneous ODE $y'' + p(t) \cdot y' + q(t) \cdot y = 0$, and u_1, u_2 are functions satisfying

$$\begin{pmatrix} y_1 & y_2 \\ y_1' & y_2' \end{pmatrix} \begin{pmatrix} u_1' \\ u_2' \end{pmatrix} = \begin{pmatrix} 0 \\ r \end{pmatrix} \tag{11}$$

And the general solution is given by

$$y(t) = (u_1(t) + c_1)y_1(t) + (u_2(t) + c_2)y_2(t) \tag{12}$$

for constants c_1, c_2.

In fact, u_1, u_2 can be given by

$$u_1(t) = -\int \frac{y_2(t)r(t)}{W(y_1,y_2)(t)}\,\mathrm{d}t \qquad \text{and} \qquad u_2(t) = \int \frac{y_1(t)r(t)}{W(y_1,y_2)(t)}\,\mathrm{d}t \tag{13}$$

Proof (Variation of Parameters)**.**

Given eq. (10) and eq. (11), we can verify that y_p is a particular solution of the non-homogeneous ODE 9. Since y_1, y_2 form a FSS of the corresponding homogeneous ODE, $W(y_1,y_2)(t) \neq 0$ for all $t \in I$ (by Theorem 1.6 and the corollary of Abel's Theorem). Thus, u_1', u_2' can be solved by Cramer's rule from eq. (11). After integration, u_1, u_2 in eq. (13) is obtained and so is the particular solution y_p in eq. (10). Finally, using Theorem 3.1, we have the general solution in eq. (12). $\square$